

Symptomatic Relief or Health Restoration? Disease Management or Cure?

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नित्यं हिताहारविहारसेवी समीक्ष्यकारी विषयेष्वसक्तः। दाता समः सत्यपरः क्षमावानाप्तोपसेवी च भवत्यरोगः॥३६॥
--- अष्टाङ्गहृदयम्, सूत्रस्थान (४.३६)

One who always takes wholesome food and recreation, acts after thoughtful consideration, is not excessively attached to sensory pleasures, is charitable, even-minded, truthful, forgiving, and keeps the company of the learned, remains free from disease. - Aṣṭāṅgahṛdayam Sūtrasthāna (4.36)

I have long been intrigued by a fundamental question. Despite extraordinary advances in biomedical research over the past century, why do we still struggle to cure most chronic diseases? In many cases, modern medicine successfully alleviates symptoms, slows disease progression, or reduces the risk of complications, but complete and lasting cures remain elusive. The prevailing biomedical approach has achieved remarkable depth in uncovering molecular mechanisms and signalling pathways associated with disease. Yet this very depth often comes at the cost of breadth. Human physiology is an intricately interconnected network, and isolating one pathway rarely captures the complexity of the whole organism. Modern medicine excels in emergency care, trauma, surgery, intensive care, and the treatment of many acute illnesses. However, when it comes to chronic diseases, treatment frequently focuses on modifying individual molecular targets rather than understanding why the body's overall physiological balance was disturbed in the first place. This is a question Āyurveda would ask.

Let's begin with the humble 'Fever'

Modern medicine possesses an impressive arsenal of drugs that can rapidly relieve pain and reduce fever. Medicines such as paracetamol and ibuprofen often bring down body temperature within hours, providing considerable comfort to the patient. When fever results from a bacterial infection, antibiotics can eliminate the causative organism, and in many situations this approach is life-saving. At first glance, it may appear that the underlying cause has been successfully treated. However, this is where a deeper question arises. Suppose a person develops pneumonia because of a bacterial infection. Modern medicine correctly identifies the bacterium and administers an appropriate antibiotic. The infection subsides and the patient recovers. But should our scientific inquiry end there? Or should we ask an additional question: Why did this particular individual develop the infection in the first place, while countless others exposed to the same microorganism remained perfectly healthy?

The human immune system is remarkably sophisticated. Every day, it encounters innumerable bacteria, viruses, fungi, and parasites, eliminating most of them long before they produce any symptoms. Disease develops only when this intricate defence system is unable to maintain its normal protective function. In that sense, the microorganism is only one part of the story. The susceptibility of the host is equally important. Antibiotics undoubtedly eliminate the immediate bacterial infection and have saved millions of lives. Yet if the underlying factors that predispose an individual to recurrent infections remain unaddressed, infections may continue to recur throughout life. Eventually, one may encounter microorganisms that are resistant to available antibiotics or infections that become increasingly difficult to treat. Thus, while antibiotics often resolve the immediate infection, they do not necessarily explain why the body's natural resistance failed in the first place. From the Āyurvedic lens, this becomes a central question: Identifying the invading pathogen isn't enough, but why was the body's innate capacity to maintain health compromised?

Tackling Diabetes

Diabetes provides another illuminating example. Modern medicine has developed an impressive array of drugs that lower blood glucose through different molecular mechanisms. Some stimulate insulin secretion, others improve insulin sensitivity, while still others reduce glucose production or increase its excretion. These therapies undoubtedly

help control blood sugar and reduce complications. Yet the fundamental question remains: Why does the body's glucose metabolism become deranged in the first place?

It is not that biomedical research ignores this question. On the contrary, enormous scientific effort has gone into uncovering the molecular basis of diabetes. However, the prevailing strategy has been to study the disease by progressively dissecting individual pathways. One abnormal molecule leads to another, one signaling pathway reveals several more, and each new discovery opens yet another layer of biological complexity. This reductionist approach has generated an enormous amount of knowledge, but often without providing a unified explanation of why the disease develops. Consider insulin. Insulin is a hormone produced by the pancreas in response to rising blood glucose levels, such as after a meal. It enables glucose (sugar) to enter the body's cells, where it is used or stored, thereby lowering the level of glucose in the bloodstream. The discovery that insulin lowers blood glucose revolutionized diabetes treatment, and insulin therapy has saved countless lives. Yet many patients with Type 2 diabetes eventually become less responsive to their own insulin, a phenomenon termed insulin resistance. This immediately raises another question: Why does insulin resistance develop? Researchers observed that obesity is frequently associated with insulin resistance, but the direction of causality remains far from straightforward. Does obesity cause insulin resistance, or does altered insulin signaling promote weight gain? Interestingly, insulin therapy itself often causes weight gain. Recent research has further complicated the picture by suggesting that chronically elevated insulin levels may themselves contribute to the development of metabolic dysfunction, independent of obesity or insulin resistance. At the same time, insulin resistance, once regarded solely as a pathological defect, is increasingly being recognized as an adaptive response that may protect cells from metabolic stress under certain conditions. The point is not that these discoveries are incorrect. Rather, each answer seems to generate several new questions. The deeper one investigates individual molecular pathways, the more intricate the network becomes. In this process, the broader question of why the body's overall metabolic equilibrium was disturbed can sometimes recede into the background.

Another example: Rheumatoid Arthritis

For many chronic diseases, modern medicine attempts to identify one key molecule, receptor, or signaling pathway responsible for the disease and then develops drugs to target it. Unfortunately, no biological pathway functions in isolation. When one pathway becomes disturbed, numerous others are usually affected as well. Rheumatoid arthritis illustrates this well. Elevated levels of TNF- α led to the development of TNF inhibitors, which have benefited many patients. However, over time, other inflammatory mediators such as IL-6 and IL-1 often become more prominent, necessitating a change in therapy to drugs targeting these molecules. This approach may control disease activity, but it rarely resolves the underlying process driving the inflammation. Moreover, TNF- α , IL-6, and IL-1 are not harmful chemicals by themselves. They are essential components of the body's normal immune system. Suppressing them inevitably interferes with normal physiology, explaining why these drugs can have serious adverse effects. Elevated levels of these molecules are better viewed as signals that something has gone wrong, rather than the primary cause of the disease. A true cure would require understanding why these inflammatory signals become persistently elevated in the first place, instead of targeting them one by one.

Who Is Responsible for Your Health?

Another challenge arises from the economics of modern drug development. Biomedical research is extraordinarily expensive and requires sustained financial investment. Pharmaceutical companies routinely invest billions of dollars in research and development, with no guarantee that any particular project will ever lead to a successful drug. To continue investing in future research, they must remain financially viable. It is therefore natural that substantial resources are directed towards areas that are scientifically tractable and commercially sustainable. Developing drugs that target specific molecules or receptors often fits this model better than attempting to discover permanent cures for complex chronic diseases, a task that is scientifically far more difficult, uncertain, and risky.

So, really no drug from any pharma giant can permanently replace healthy living. No physician can live your life for you. No hospital can preserve your health every day. The lure of quick fixes through pills, shots, and drugs can never substitute for understanding and cultivating health. Protecting and enhancing our health is ultimately our own responsibility. We can either invest in understanding the science of health and life, or postpone that responsibility until disease compels us to seek treatment. Ayurveda teaches us that the wiser choice is the former one.